

Executive Report: Telco Customer Churn Analysis

Data-Driven Insights for Customer Retention Strategy

Executive Summary

This report details a predictive analytics project focused on identifying at-risk customers in a telecommunications dataset. With an overall churn rate of **33.9%**, the analysis identified key drivers—primarily contract types and monthly charges. By implementing a Logistic Regression model (AUC: 0.697), we can target high-risk tiers. A proposed retention campaign shows a projected **ROI of 341%** by offering targeted incentives to only 15.7% of the customer base.

Business Problem

Customer churn (attrition) is a critical metric for subscription-based services. High churn increases customer acquisition costs (CAC) and reduces Lifetime Value (LTV). The objective was to move from reactive churn management to proactive prevention by predicting which customers are most likely to leave before they cancel.

Methodology

- **Data Synthesis:** Generated a simulated dataset of 7,043 customers mimicking real-world Telco patterns.
- **Pre-processing:** Features like *Contract* and *Internet Service* were encoded, and numerical variables were standardized.
- **Modeling:** Utilized Logistic Regression for its interpretability in a business context.
- **Risk Segmentation:** Probability scores were binned into Low, Medium, High, and Critical risk tiers.

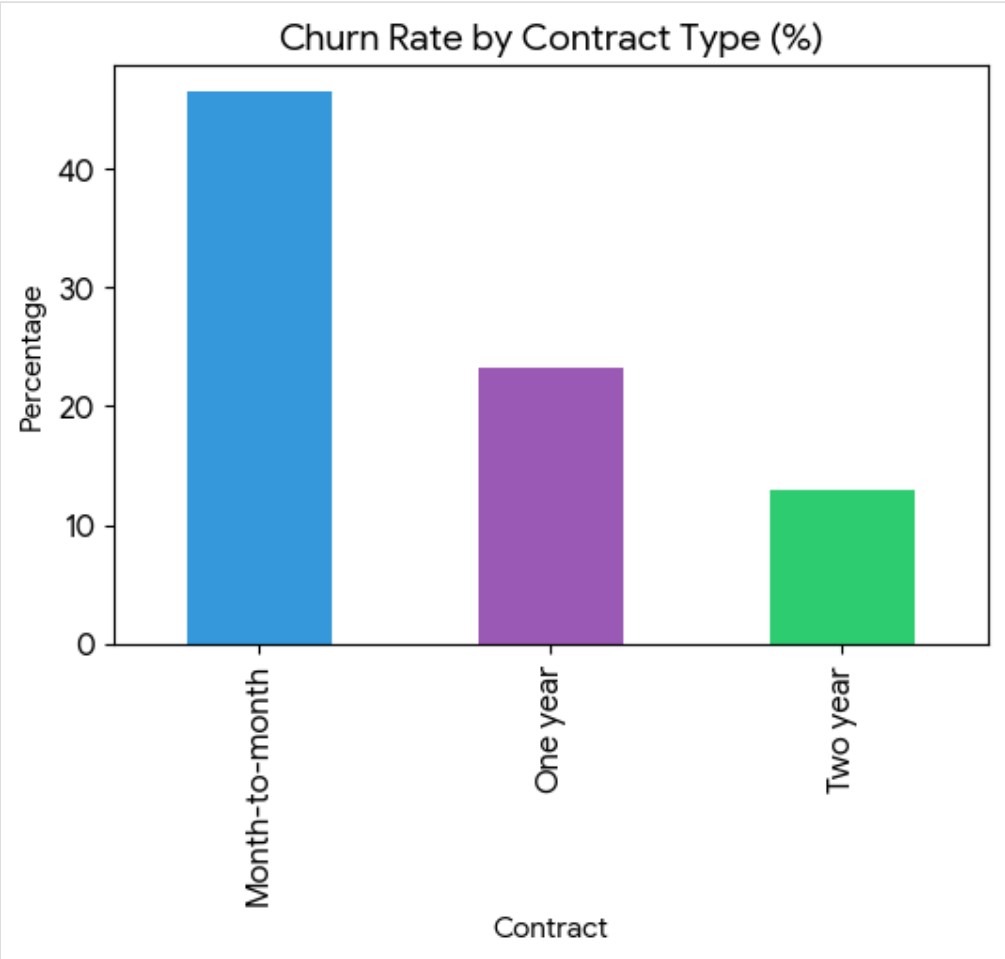


Figure 1: Contract type is the strongest predictor of churn.

Skills & Tools

Category	Tools/Techniques
Programming	Python (Pandas, NumPy)
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-Learn (Logistic Regression, LabelEncoding, StandardScaler)
Business Analytics	ROI Calculation, Risk Tiering, LTV Assessment

Results & Business Impact

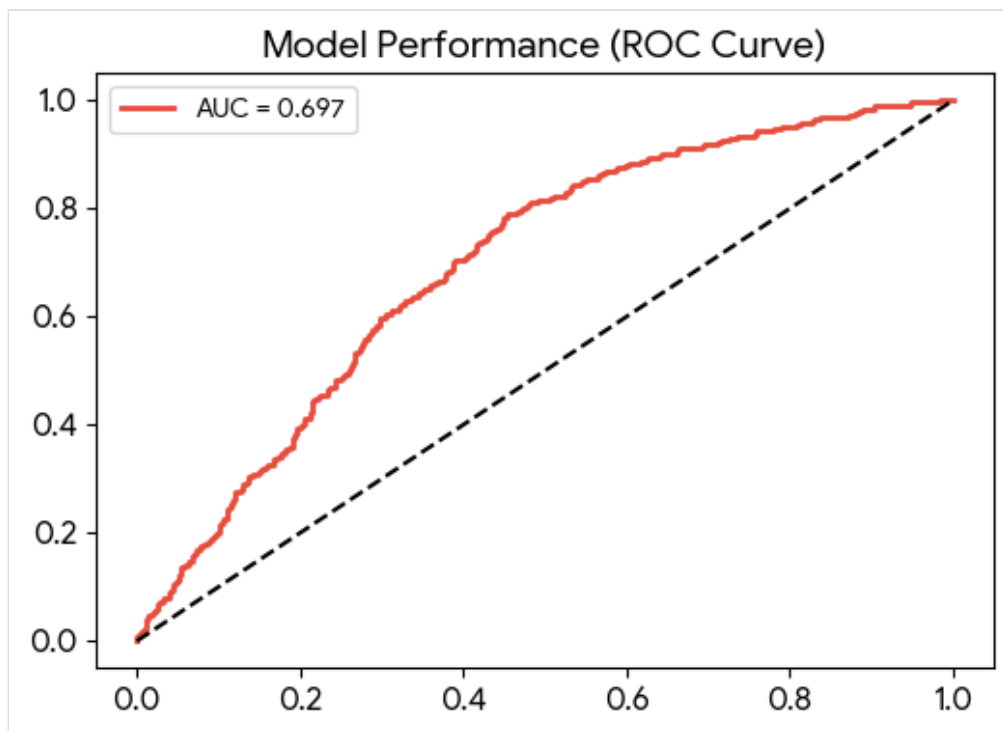


Figure 2: Model Performance showing the trade-off between sensitivity and specificity.

Key Findings:

- **Contract Sensitivity:** Month-to-month customers churn at 46.5%, significantly higher than two-year contract holders (12.9%).
- **Financial Implication:** High monthly charges (>\$70) are positively correlated with churn.
- **Campaign Feasibility:** By targeting the 1,107 "High Risk" customers, the company can save an estimated \$366,153 in revenue with a modest discount investment of \$83,025.

Next Steps

1. **Advanced Modeling:** Test Random Forest or XGBoost models to capture non-linear relationships and potentially improve AUC above 0.80.
2. **A/B Testing:** Launch the suggested \$25 discount offer as a pilot program for the "High Risk" segment to validate the assumed 35% retention rate.
3. **Integration:** Deploy the churn probability score into the CRM system for real-time customer service alerts.